

Active Inference Math Session with Octopus ~

August 20, 2024

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Duration: 44:57 | Uploaded: Unknown Transcript method: auto_caption

good too so here's where uh we saw the these are the stochastic differential equations and what this says is the difference in U is going to be some function of the current XU and Y and if that's all we had if this is all we had then we would have a differential equation this would say the update is given by the change is S_u is determined by XU and Y and what the specific rule for determining the change is can be you know up for discussion but the we're describing the change through a rule that depends on the current information what makes it a stochastic differential equation is that it has this additional $\Omega_{sub U}$ term um where $\Omega_{sub U}$ is a random variable so we're not always going to get exactly the same change based on the current information but rather we will get a rule that is applied and then some initial stochastic noise is added on so we will not just will not generate a single trajectory like an ordinary differential equation would um rather it's going to generate a bunch of different possible trajectories and each of these different possible trajectories will have a probability associated with it and that's going to be dependent on what the original uh deterministic update rule $f_{sub U}$ is and what the original stochastic process $\Omega_{sub U}$ is so this is you know incredibly General uh and if we plugged in specific f_s and specific $\Omega_{sub U}$ then we might see specific results but this is an example of something that we talked about a little bit last week which is a stochastic differential equation and the point here I don't have tremendous like I've never taught a class on these um I've seen them a little bit in my work on cryptocurrency Consulting Financial modeling um but mainly I know this from seeing other people do it and so this is something I'm actively working to try to understand better so one thing that I find is helpful is to see if there is a specific example of this that I can come up with so what would it look like if I um came up with like what is a stochastic process do I know an example of a stochastic process do I know uh so one of the the PE things you can do here is actually done in the book you can start removing information and seeing what pieces you actually have because for me as someone with my brain um I get overwhelmed by information when it occurs in like large

amorphous chunks and so what I have to try to do is give it a structure and I liked how the book did this with the expected free energy this has five parts right but then you start looking at what if I drop this part what if I drop this part what if I drop these two parts and maybe I understand this part where I just dropped two it's it's uh I don't have a name for it it's like you cover up part of the information and you see if you can grasp that single piece so I do have a large principle for this I've developed which is called the two confusion principle or the um stepladder princip principle and it says try not to be try to go one step at a time and you'll call it the single uncertainty principle or step ladder principle so when you're trying to climb these things resist the temptation to like jump four steps at once go one step at a time find one thing you are uncertain about and fill in the one fill in that one step so I developed this principle when I was trying to teach machine learning or when I was teaching machine learning and I was explaining uh or we were discussing different models like we're discussing a random Forest we're discussing a boosted tree whatever and then we're looking at data we're trying to look at data on um stock prices well there's a problem here because if we don't if we don't know much about stock prices and we don't know much about random forests we have almost no hope of understanding anything because we don't have a starting point we don't have a concrete way of like how if I understand one I can use it to understand the other so this is a principle I've been trying to use I understand differential equations don't understand stochastic differential equations so I'm trying to leverage whatever I do understand about the deterministic case to advance to that next step of my understanding and a lot of times just stopping and saying okay what am I confused about what what of this do I not understand um there's something psychological that happens here because if we don't have any uh contact with the familiar at all our brains tend to go into modes of uncertainty anxiety and then we it's easy it's easy when we get in this uncertain place for the uncertainties to begin to spiral that's something I don't know if you all have experienced it's not pleasant but like when you're just imagining things you can start to imagine basically anything whereas if you make contact with some aspect of concrete reality then suddenly you have a grounding point so are there any things that we're that we're seeing where maybe we could find like the next step of what we would want to fill in or where we're confident up to a certain point and it's okay one of the things for me at least it's okay you have no idea how deep these things could go none of us do so we got a few weeks ago we were looking at the different base of the logarithms and how that could change the answer we got if we're trying to come to consensus on what an answer should be but we're using different bases of logarithms you know that's that's a place where we

could need to fill something in the key though all of this that I'm trying to do is not just okay let's review these symbolic manipulations or let's prove this theorem although those things might be important but there's also this highly metacognitive strategic aspect of how do we decide what we're going to think about and how we're going to think about it to leverage and uh I have all these things I've developed this this toolkit for getting to math mathematical understanding from wherever we're starting and I'm trying to share those tools as well as hopefully using them to to gain some insight so I'm be quiet for a few minutes and see if you all have any thoughts on what I've said or any questions as to specific things you'd like to dive into e e boiler's number and when you would use the natural logarithm versus other logarithms and why yeah why would you use why would you use a particular base for a particular logarithm um Oilers I mean it's called the natural algorithm it's kind of a copout but it's the rate at which things tend to grow naturally um there's also it's also useful because e to the x is its own derivative and so sometimes that cleans things up where you don't need an extra constant hanging around um so let me talk about why would you use log base 2 why why would you want to use two for the logarithm logarithmic base so computer scientists are going to use log base 2 because they're working with switches right and things are either on or off and so it makes sense in terms of the kind of also they're often working with trees to model that so log base 2 comes from things tending to grow like binary trees and sort of like when you get to the next level there's twice as much so using log base too kind of helps you know what level you are on in this in a binary growth process roughly speaking um so why would you use log base 10 what's the advantage or what intuition comes from using why would we use log base 10 we have 10 fingers and so the metric system is based um so are you working at the order are we looking at kilometers or are we looking at centimeters or millimeters log base 10 does a very a good job of telling us immediately kind of what scale we're on where scale is a metric system type scale or number of digits in the number here we're looking at number of bits what's the advantage of log base e well not sure I've ever thought about that before thought way more about the other two uh types of logarithms and even though as a mathematician I use log log base e all the time uh I'm not sure why that's the my go-to it has these nice properties there are some nice properties about e to the X and e as a number some limiting things um like if you are doing continuous continuous growth continuous compounding e kind of naturally pops out but as to what intuition it gives in terms of the I don't have intuition for it the same way I do for what log base 2 and log base 10 tell me so maybe that's an interesting thing we could um put in our sack to think about for the future of

how like how intuitively does log Bas E help us and if anybody has thought on that I'm happy to consider them put that in the it's usually just important that once we pick a base we're consistent with it or that we are clear on what base we picked but I'm sure there is some intuition here um another technique that's useful here or has been useful to me at least in terms of keeping things approachable and concrete just like active inference with these mathematical things a lot of times there's a high road and a low road and or maybe we can think of it as abstract and concrete so with all of these equations you know maybe some people have sufficient experience that they see it and it immediately means something to them and there's really no need for them to do else I don't do anything else I don't you know I can't be inside other people's brains and know how they're thinking um I do know that a difference between mathematical thinking and other types of thinking in my opinion is that there's really less discrepancy between what is said and what is represented and what I mean by that is in like political speech you can talk about all these very abstract Concepts like equality or Freedom or whatever you can come up with you know very elaborate and nice sounding phrases but they lack any real reference so when we come to what does freedom actually mean what does freedom look like then it becomes its own huge discussion that is ultimately unresolvable and math doesn't work that way when you say something mathematically it's intended to have a very precise meaning and to be unambiguous and uncontroversial as to what the term is representing so one thing that I has been helpful to me in looking at these things is to look for running examples or for things I can plug into the theory to sort of see how it would work so one um thing with the with these different discussions of variational free energy and expected free energy is you can just think about a common situation where you would have two different probability distributions and and then calculate and see what would the difference be or what would these quantities be for these two different situations that you could consider so as we use it over and over again it can become more concrete to us um so a couple of weeks ago something we started to do and uh did not did not arrive at but we could we could go back to it is looking at this for two different coins if you think that one coin if you think that you have a 50/50 coin but you actually have a biased coin how much uh what do these things look like what does cobac Liber Divergence look like and that was the one we were actually looking at so if we're approximating a a biased coin with a Fair coin how do these numbers change as the bias gets more and more away from 50/50 whereas the distribution gets further and further from um uh equal on each outcome so I think it would be worth going back to that if you have any interest in doing it hello we have a newcomer how are you cool so I'll be back

in just a moment and we can go through the different um we can go through koback lebl
 Divergence for two coins and then we maybe we could look at other models and and see what
 that looks like okay so let's define our two distributions here we're going to have two different
 coins and let's have P be the fair one and U let's have actually let's have P be the bias one and
 let's have q be the fair it's thicker so Q is going to be the biased coin and let's take this and so p
 is distribution Where Heads has some prob heads has a probability of uh 5050 and Tails has a
 probability of 5050 and Q I don't know what the probability of H is so let's just say that it's the
 number B and then the probability of tails is 1 minus B and I'm using B here to stand for bias
 so let's I'm going to use the second formulation [Music] for computing F of q y and I'm mainly
 going to be focused on the What's called the complexity term here so the complexity this d sub
 KL Q ofx double line P of X is the cback Leer Divergence of this so my memory serves this is
 equal to the sum over our possible outcomes let me just look it up since I'm not remembering
 it immediately one thing that's fun is the way the Q and the p in the book are uh reversed from
 the Q and the p and the formula so I have to make that mental switch um so this is going to be
 the sum over Q ofx sum overx and X of Q ofx times the log of Q ofx over p ofx and so this
 sum over x and x is like the H and the T so we're going to have an H term and a t term so we
 have q of H log Q of H over P of h plus q t log e U of T over h of T so this will be so Q of H
 was whatever my bias is I don't it's a value it's a number B okay so then log B over 0.5 Plus 1 -
 B log of 1us 0.5 so this depends on B but we can now um that's all it depends on is the
 probability of heads so we can make a plot of this where we have this quantity is the output D
 is the output and it's its function of B and it will be equal to this and what this could give us is
 an intuitive sense of for instance how quickly the coolback leeler Divergence grows when as
 the bias gets further away from 1/2 as the coin becomes less and less Fair so see if I can plot
 that with I don't think I can plot it with D's and B's and Desmos but let's try anyway and if I
 can't we'll just switch the variables to F and X for the standard D of B equal to B time log B /
 0.5 1 minus B times the log ofus B desired by 0 2 five this is what it looks like it's roughly no
 it's concave up symmetric about 0.5 maximum value for this base a loger them look it's have
 to take a limit looks like it's approaching 0.29 maybe 0.3 what this tells us is this quantity of
 cack leeler Divergence um between these two which is also called complexity then we can
 look at the accuracy the accuracy depends on data y and for me I feel like okay if I did this and
 I looked at this and I now know that this quantity looks like this for two coins I can understand
 intuitively if I have a biased coin but I think it's fair um the complexity gets higher and higher
 as the coin moves further and further away and I can can kind of see the shape of how that

happens and so that's something that's helpful to me and I try to I'm trying to do that with all of these things to come up with some simple concrete calculation I can do that connects to something I feel I understand so that it's not just a huge pile of symbols and then I've also noticed in looking at the appendix that this there are times where this pops up as a actual calculation in a more relevant context for instance where people are like let's let's find the complexity of using one distribution to approximate another or let's calculate something about that so having done this once I feel more prepared to do it again and that is called the poia principle it's named after um George POA who wrote how to solve it he said if there's a problem you don't feel comfortable with there's a smaller problem that's related that you do feel comfortable with it and you can solve that one instead as a starting point in this case the problem we worked through is like just finding an immediate concrete example to try to latch on to but that's the whole strategy is to create things in chunks so that it never becomes totally overwhelming okay is there anything y'all would like to do that with for next time uh that's currently in the textbook where you'd like for us to try to go through it in in a concrete way and make it more um what's the word less abstract less vague hello sorry I uh do can you hear me I can hear you yes yes uh thank you for doing this I'm um yeah I was a bit late so I didn't hear what you said in the beginning and it's the first time I'm here uh so maybe I'm saying things that are redundant with what you've done before so uh but um for me it was especially chapter 4 which was very unclear how things uh well kept together somehow like there were some jumps in the it's a while ago that I looked at it so I would have to look again to be very precise in what I don't um but also in this chapter the um the three formulations of the the free energy uh how one goes to another uh is totally unclear to me like how the these derivations uh how that they actually are the same that was it wasn't it wasn't clear to me at first either actually last week was unclear but I guess yeah last so we this is like the third or fourth of these sessions we've done and we're still trying we're still working through exactly what a structure or format looks like so um I'm just trying to go through things that might be helpful both in terms of if people have specific questions I'll do what I can to figure out where the information is or what we can do with it and then also is I'm I'm not um I have a strong math background but I don't have a strong background in all of the math in the active inference and I I'm learning active inference so a lot of times it's like oh okay I read this this week and here's how I tried to understand it just sharing things that worked for me or might might work for other people our goal is to get is to unstuck ourselves as much as we can and just have a richer exper experience with it the the specific thing you're talking about with the proofs of the

equivalence of the different types of variational free or the different calculations of variational free energy we actually did that last week in detail right I can share that link uh with you and then yeah which will be very nice it from there um are you on the Discord I can put in the Discord and I think you can uh is there a Discord I not on the Discord I am on the Koda ah okay I will add it to the Koda then a link to it on the Koda and um I have my own page which I need to start using more for these discs and uh but I I'm not very acquainted with Koda yet so um yes I'll try to find it and then let me let me see because uh yeah I'll try to find it too and um if you come back next week I can be sure to have the link ready for you okay thank you very much wa I'm going to check to be certain that I can be there next week otherwise it's the week after I guess yeah uh yeah that should be fine next week I'll try to be there cool um so in the Cod is there a Discord that you said uh yeah there is a Discord let me see if I can drop I'll drop you an invite via private message in the chat thank you cool well uh if you don't mind my asking how do you pronounce your name friend uh my name is yoris yoris yoris Heros okay okay yours is that I hope yes that's fine have heard many many variants by now so it's fine whatever I'll do my best um youris um yeah so I was actually starting this session by saying I was low on Preparation like last week I had all the proofs written out and some exercises we were going to do together and this week it just didn't happen that way but if people can let me know specific questions ahead of time I usually try to prepare something and if not we just kind of show up and and freestyle it because okay great anyone have any other thoughts they'd like to share before we close okay well I am getting the Discord invite I like your shirt oh thank you and thank you for today yeah thank you all if we keep coming we'll keep we'll keep learning yeah so if anyone is not in the Discord and you would like to be I'm putting a link in the chat right now and I hope you all will come again next week and if you let me know questions in the Discord I'll try to think about more specific things that we could discuss thank you very much see you next week then see next week